

2023-Ph-H1-Q15

Section: Particles and Waves

Topic: The Inverse Square Law

A radioactive point source is moved from 0.5 m to 1.0 m away. What happens to the intensity?

Answer: C

(It becomes one quarter)

Guidance for Students:

Inverse square law:

$$I \propto \frac{1}{r^2}$$

If distance doubles, intensity becomes $\frac{1}{2^2} = \frac{1}{4}$

Revision Tips:

- $I = \frac{P}{4\pi r^2}$
 - Doubling distance $\rightarrow$ intensity $\div 4$
 - Used in light, sound, and radiation problems
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